

Eñica moderna

Eñica cuntica

$$E = h \cdot f$$

$$f = \frac{c}{\lambda}$$

$$E_{inc} = W_{ext} + E_c \begin{cases} \rightarrow E_c = \frac{1}{2} m v^2 \\ \rightarrow W_{ext} = h \cdot f_0 \\ \rightarrow E_{inc} = h \cdot f \end{cases}$$

Potencial de frenada

$$q \Delta V = \frac{1}{2} m v^2$$

$$V_{frenada} = \frac{E_c}{|q|e}$$

$$\lambda = \frac{h}{p} = \frac{h}{m \cdot v}$$

$$\Delta x \cdot \Delta p \geq \frac{h}{4\pi} \rightarrow \Delta x \cdot \Delta v \cdot m \geq \frac{h}{4\pi}$$

Eñica nuclear

$$T_{1/2} = \frac{\ln 2}{\lambda}$$

Periodo de semi-
desintegracin
↑ vida media

$$N = N_0 e^{-\lambda t}$$

↑ n de ncleos sin desintegrar

1 Bq → 1 desintegracin/s

$$m = m_0 e^{-\lambda t}$$

$$A = A_0 e^{-\lambda t}$$

↑ actividad: n de desintegraciones/s → $A_0 = N_0 \lambda$
 $A = N \lambda$

$$\Delta E = \Delta m \cdot c^2$$

↑ defecte
de masa

$$\Delta m = m_{reactivos} - m_{productos}$$

$$\Delta E_m = \frac{\Delta E}{A}$$

$$\Delta E = A \cdot m \cdot c^2 \cdot \Delta m [2m_p + (A-2)m_n] - m_m$$

